

Medical Command Center Under the Hood

How every screen, agent, and guardrail actually works

Detailed flow diagrams and infographics covering claim intake, the grounded AI assistant, Growth Studio's process-modernization tooling, and the technology underneath all of it.

1. System Architecture	2
2. Claim Intake — Detailed Flow	3
3. Manager Review & Audit Flow	6
4. AI Claim Manager — Chat & Voice Flow	7
5. Growth Studio — Discovery to Revenue Flow	9
6. Dashboard & Reporting	11
7. Technology Stack	12
8. Guardrails Reference	13
9. Status & Badge Glossary	14



1. System Architecture

Five layers, each independently swappable — the pattern used throughout the app: a model/engine drafts or computes, and the layer above it never blindly trusts what came from below without a code-level check.

Client Layer

Angular 17 (standalone components) · Web Speech API (voice input) · Web Audio API (real-time lip-sync analysis)

↓ HTTPS / JSON ↓

API Layer — FastAPI routers

documents · claims · chat · dashboard · tts · discovery · simulation · roi · revenue

↓

Service & Agent Layer

PDF/OCR extraction · relevance scoring · claim field extraction · chat grounding · discovery · agent simulation · ROI & revenue engines

↓

AI Provider Layer (swappable)

Google Gemini (configured) ⇔ deterministic rule-based provider (zero-credential fallback, always available)

↓

Data Layer

In-memory repositories (claims · upload jobs · discovery assessments · chat history) — repository pattern, ready to swap for a real database

Why this shape matters

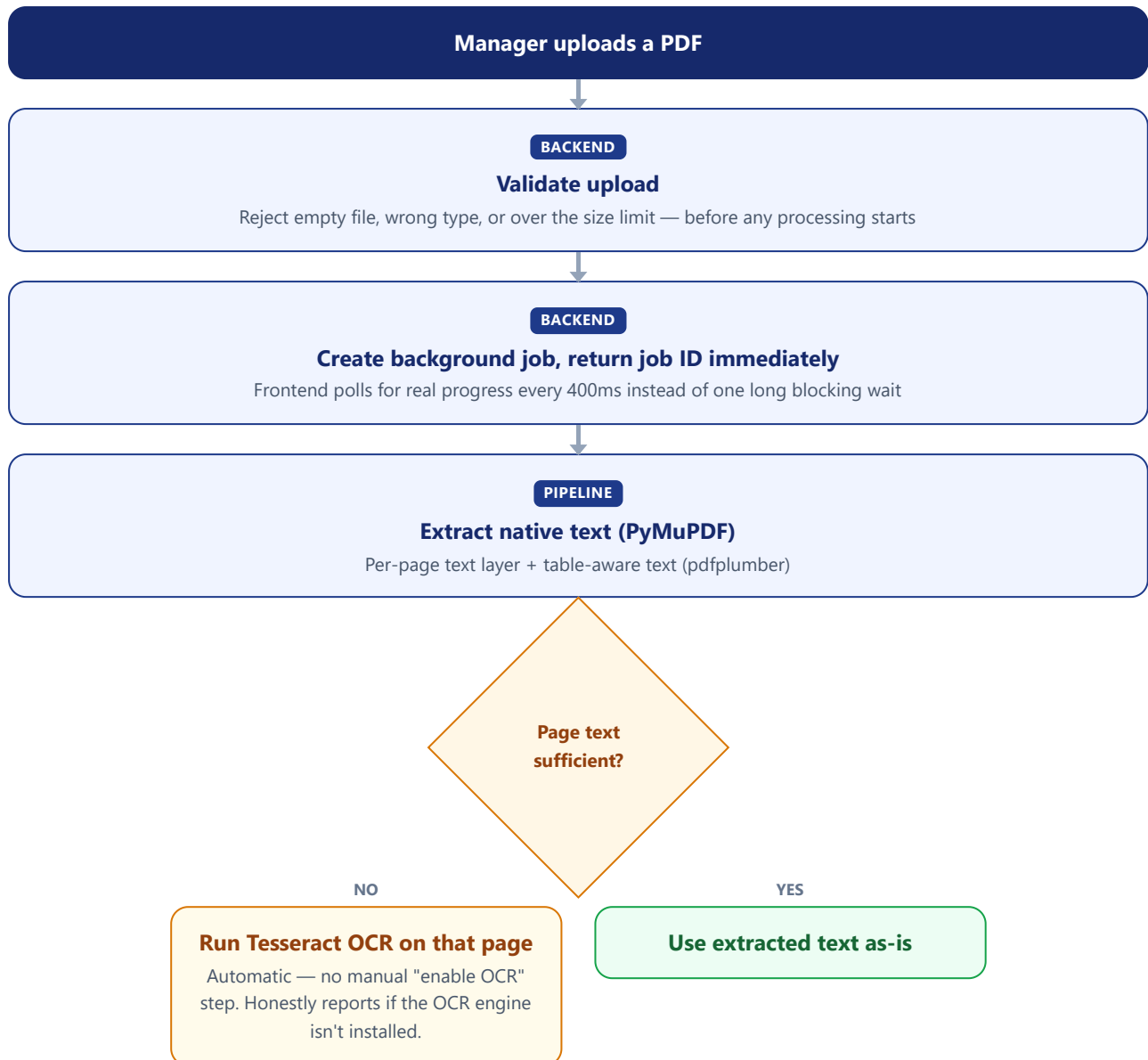
The AI never has the final word

Every AI output that affects a compliance decision (extracted field values, exception flags, agent whitelisting) passes through a code-level check in the Service layer before it reaches the API.

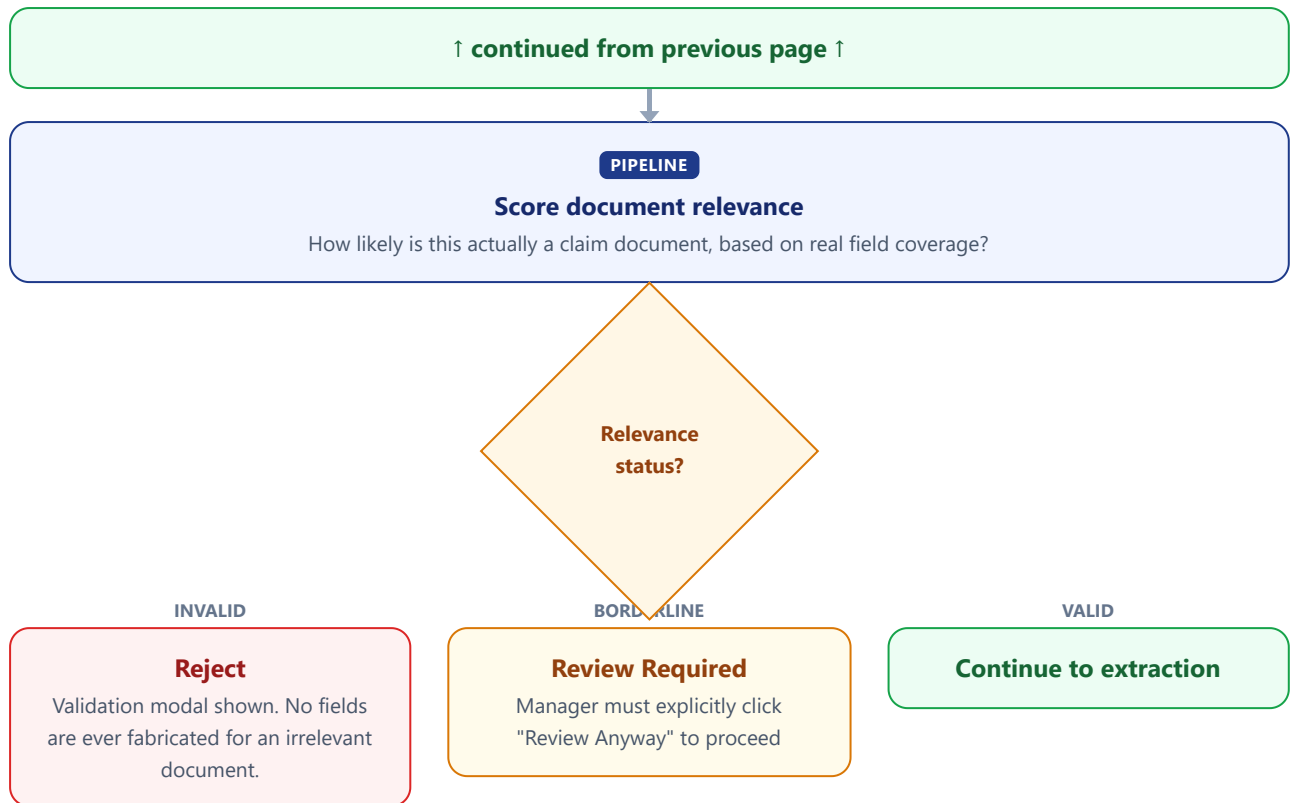
The app runs with zero AI configured

The rule-based provider isn't a stub — it's a real, working fallback for extraction, chat, discovery, and simulation, so a missing/rate-limited API key degrades the experience, never breaks it.

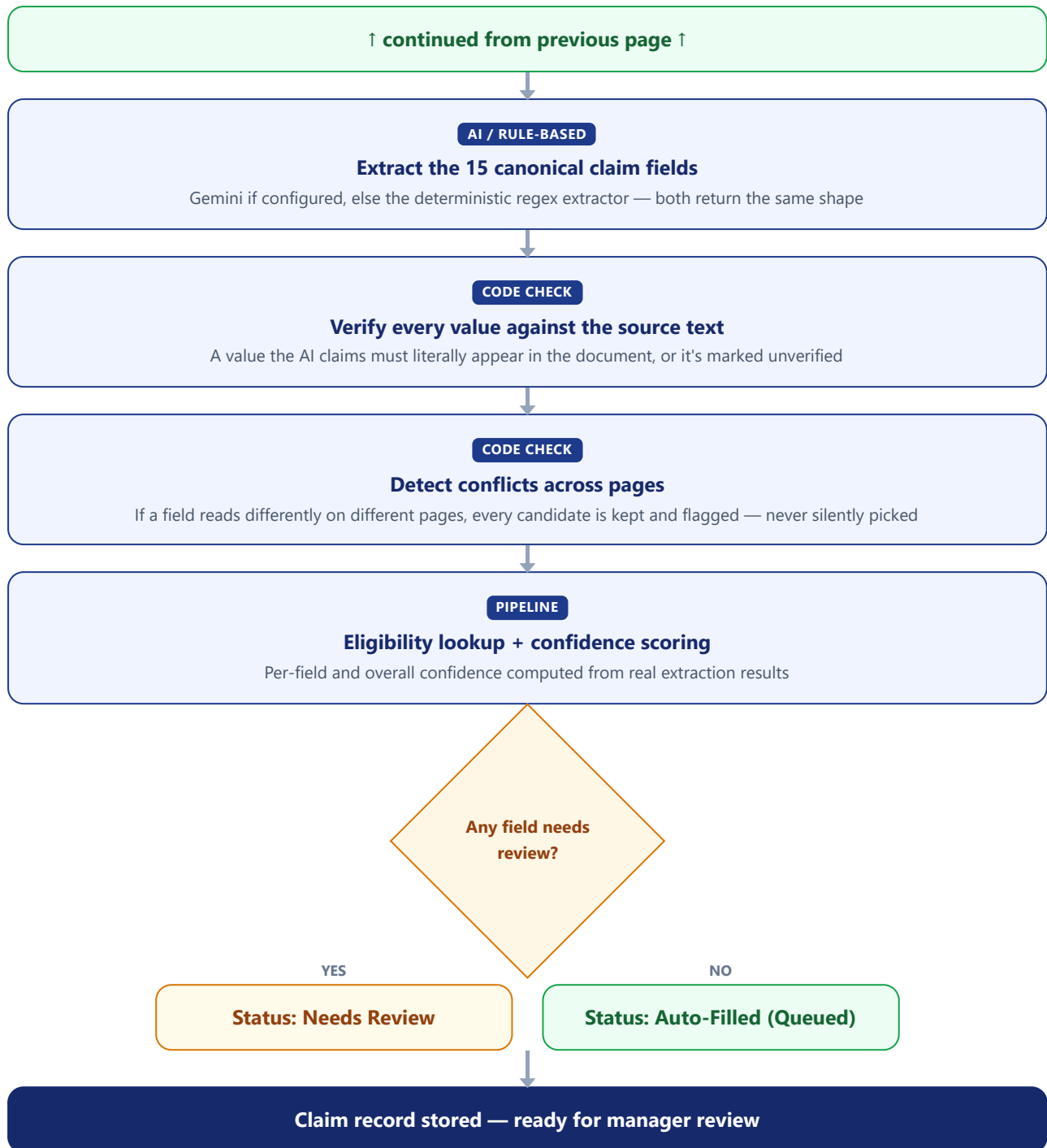
2. Claim Intake — Detailed Flow (1 of 3)



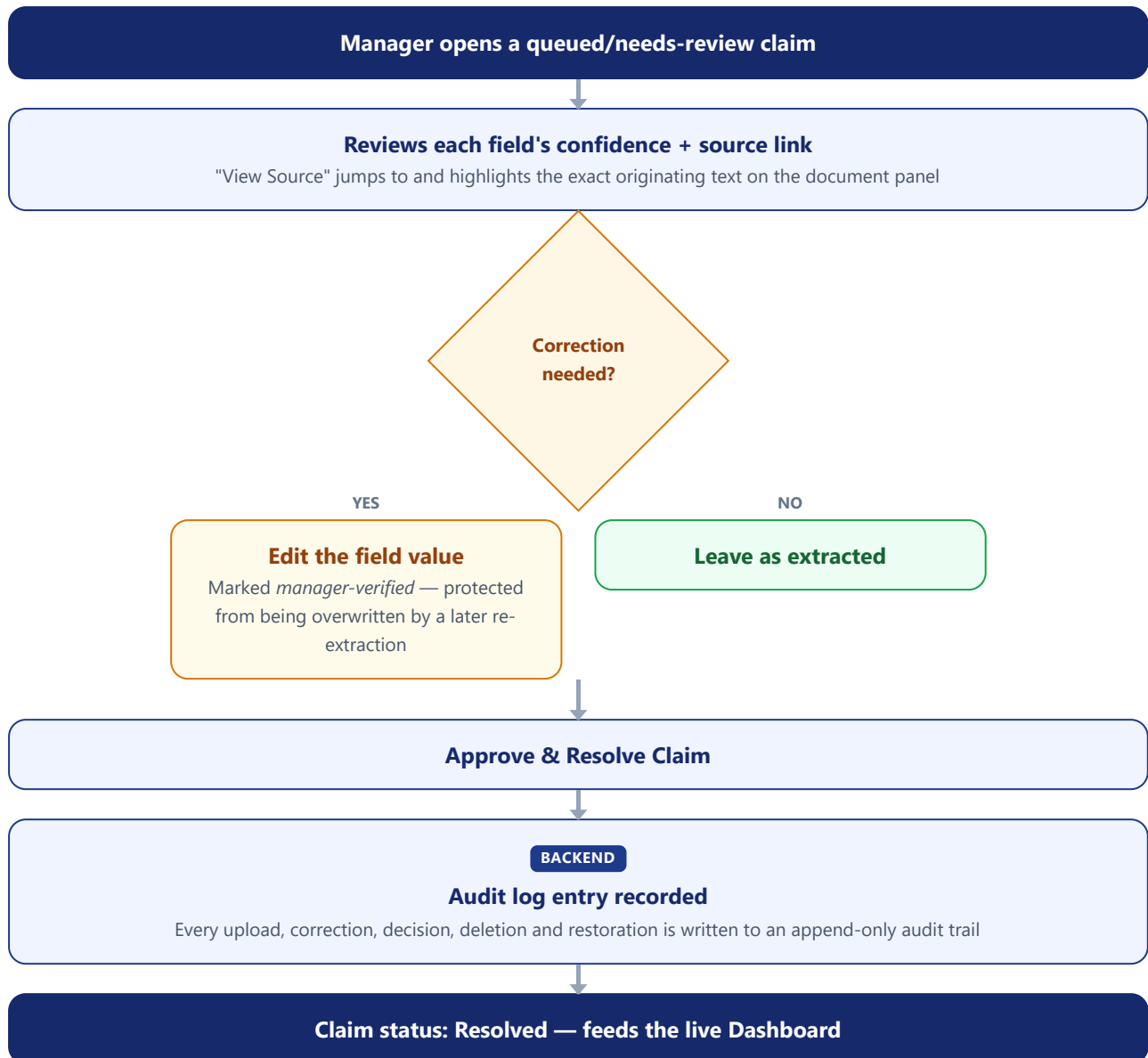
2. Claim Intake — Detailed Flow (2 of 3)



2. Claim Intake — Detailed Flow (3 of 3)



3. Manager Review & Audit Flow



Also always available

Delete / Restore

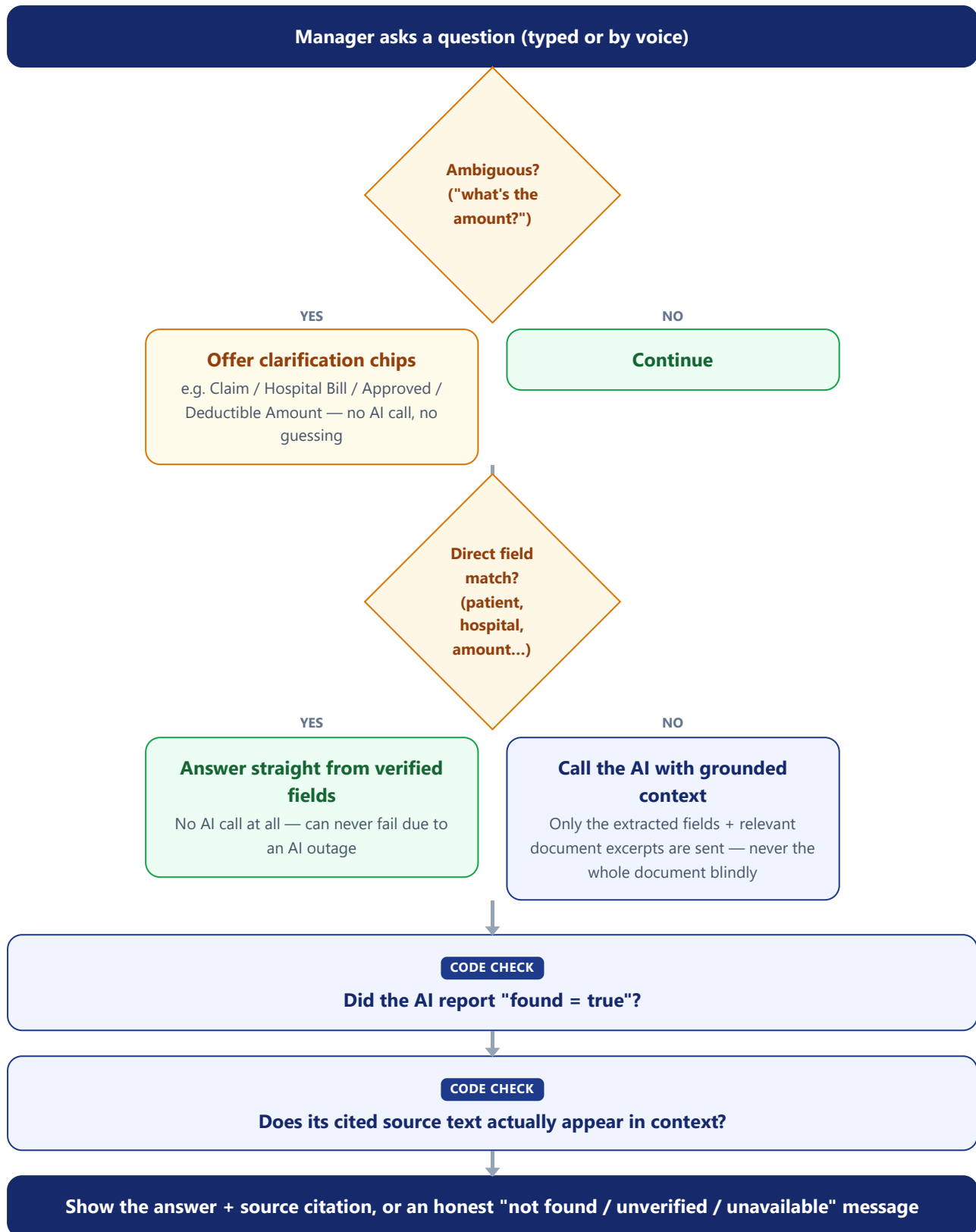
Soft delete only — nothing is destroyed. A deleted claim moves to the Deleted view and can be restored at any time.

Re-run Extraction

Re-processes the already-extracted text through the AI/rule-based pipeline again — manager-verified fields are always preserved, never overwritten.

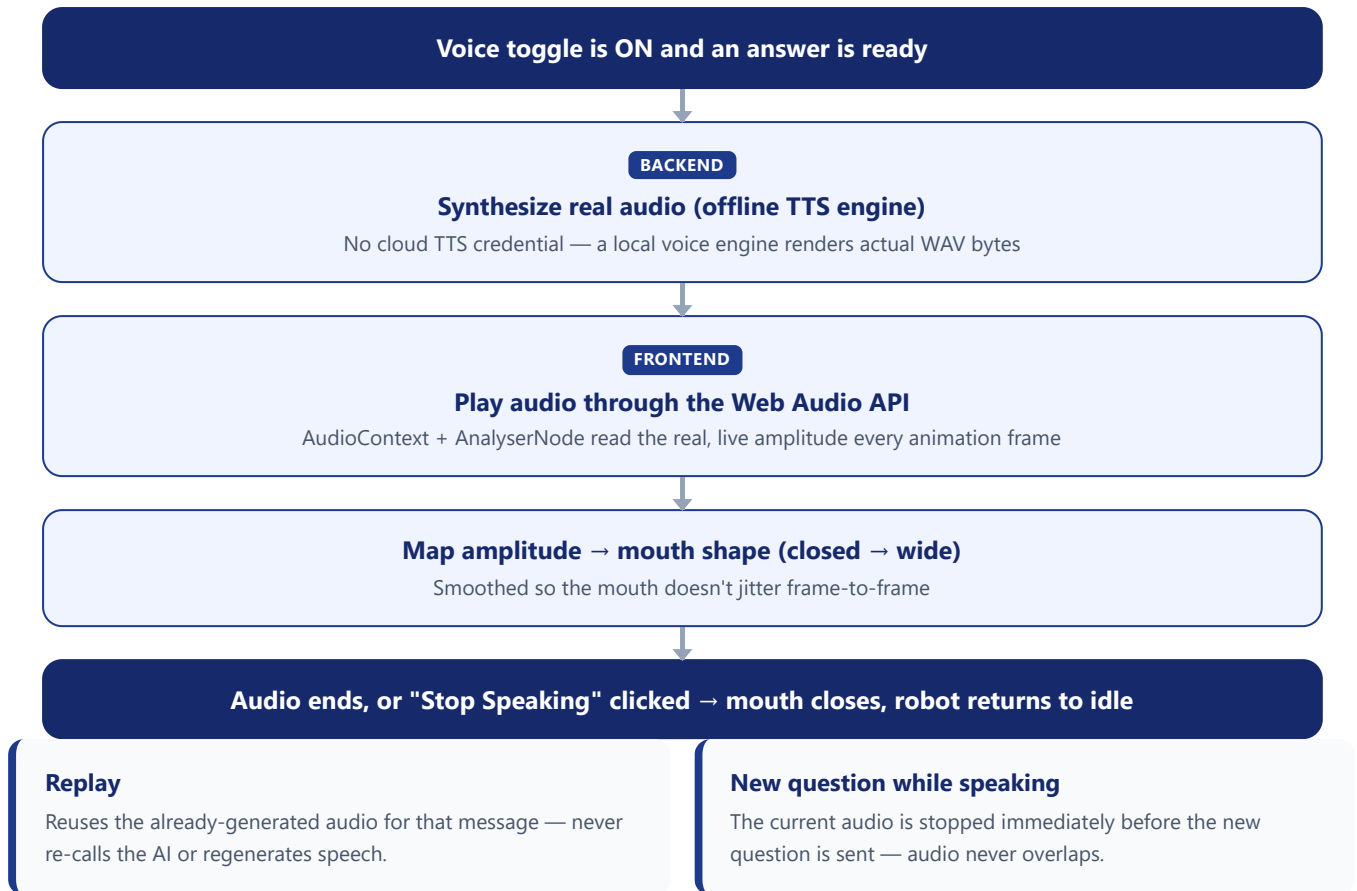
4. AI Claim Manager — Chat & Voice Flow

Grounding is enforced in this order — each step is a real code check, not a prompt instruction.

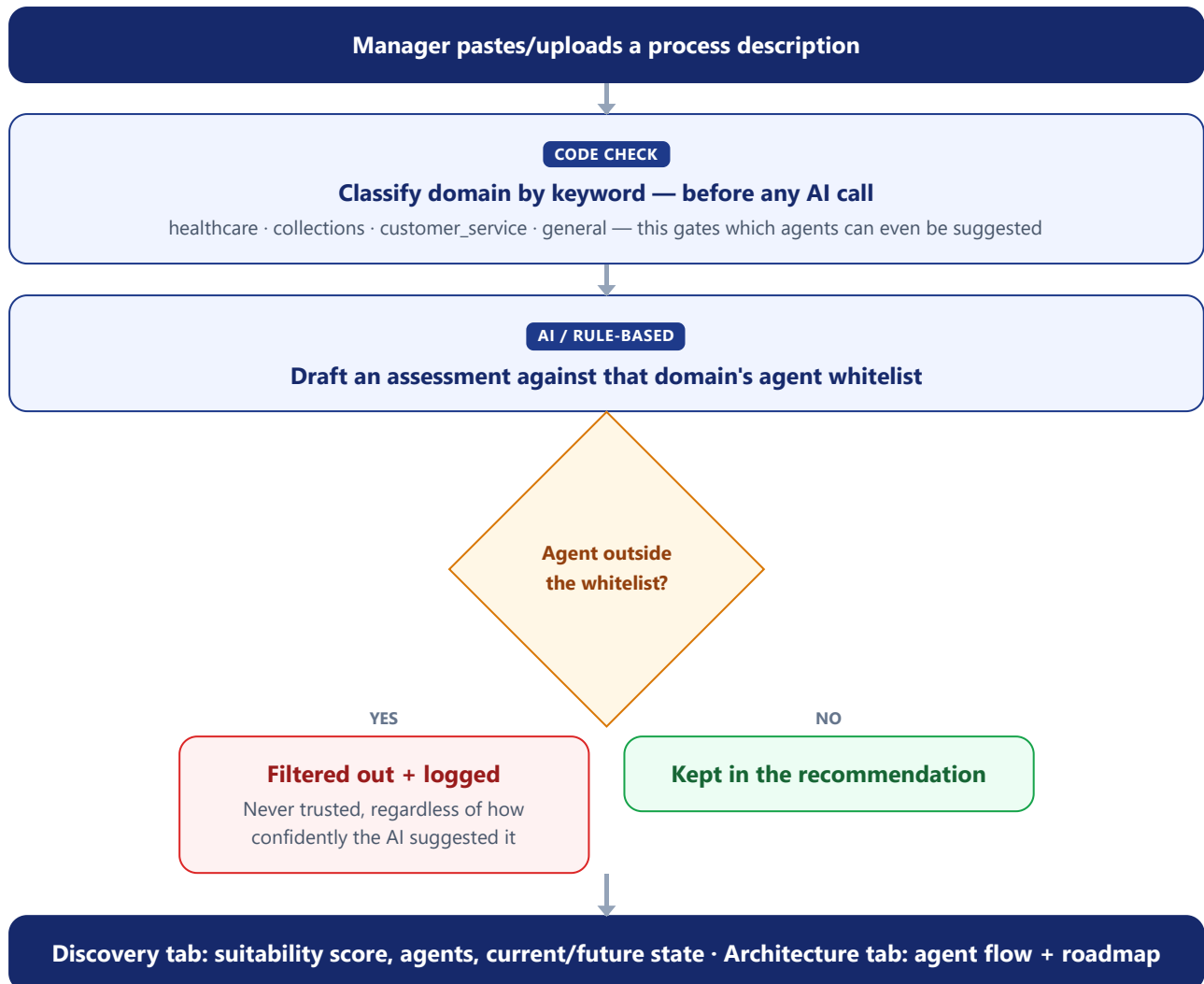


4b. Voice Output — Real Audio-Driven Lip Sync

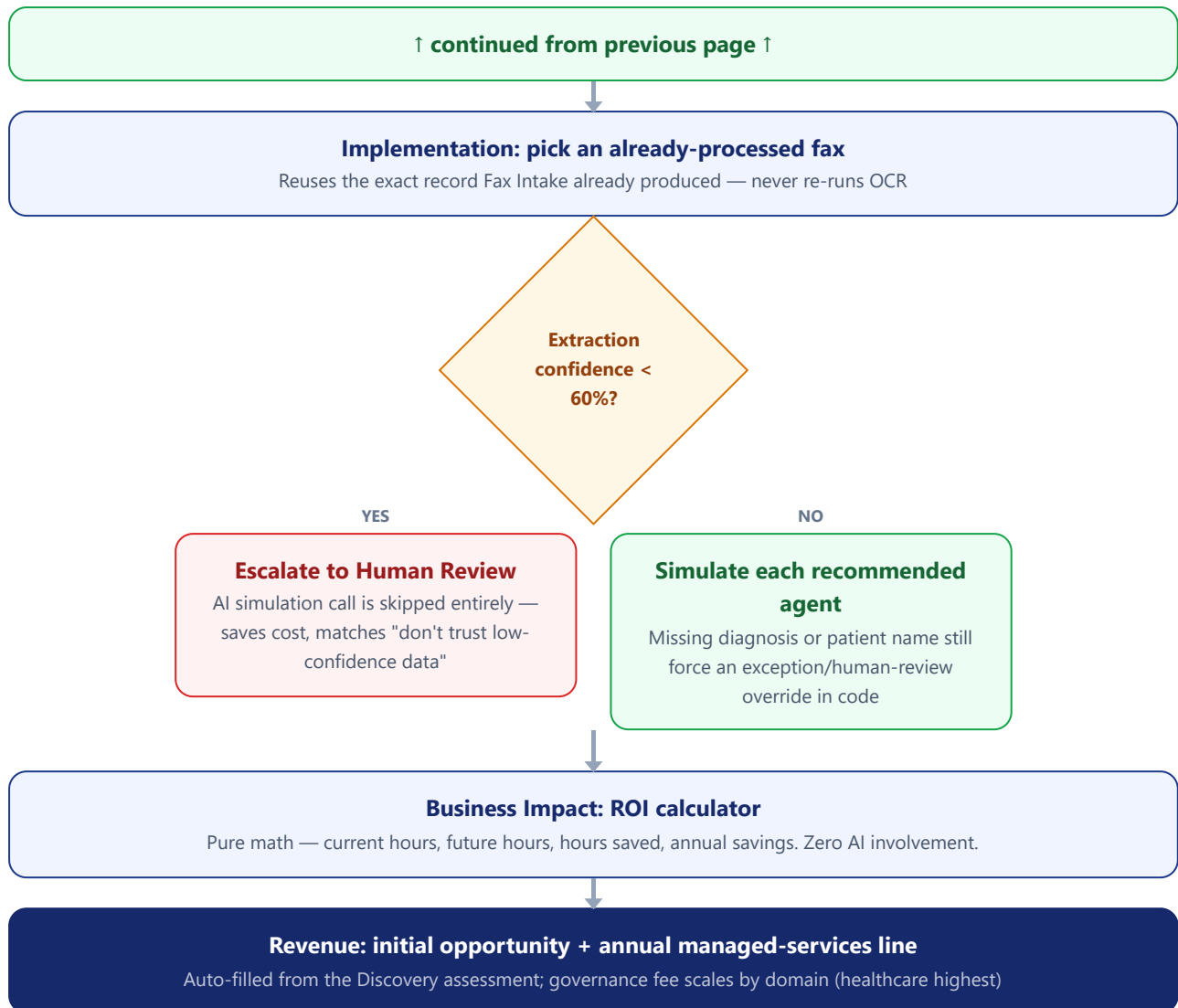
The robot avatar's mouth is driven by the actual audio signal, never a looping animation.



5. Growth Studio — Discovery to Revenue (1 of 2)

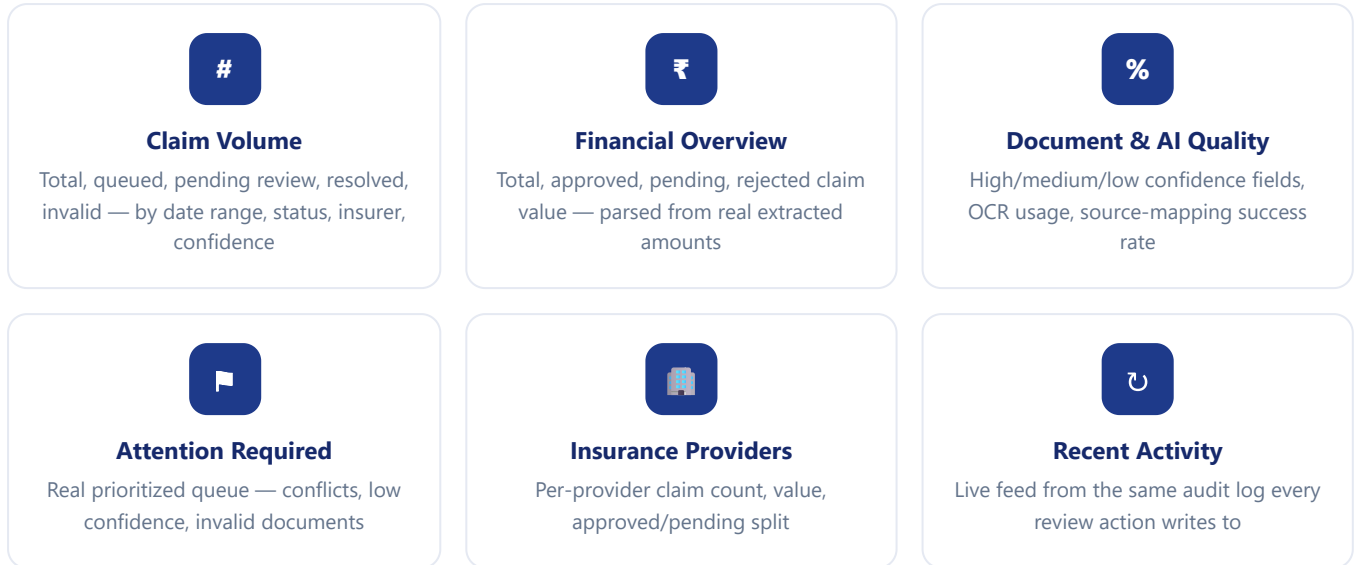


5. Growth Studio — Discovery to Revenue (2 of 2)



6. Dashboard & Reporting

Every number on the dashboard is computed live from the claim database for the currently selected filters — nothing is a placeholder or a simulated trend.



Header-level tools

Global Search

Filters the real claim list by ID, filename, or patient name — no fabricated results.

Notification Bell

Backed by the same "needs attention" logic as the dashboard queue — a real count, not a decorative badge.

7. Technology Stack

A**Angular 17**

Standalone components, client-side routing, light/dark theme

F**FastAPI**

Layered routers → services → repositories

G**Google Gemini**

Configurable AI provider, JSON-mode structured output

R**Rule-Based Fallback**

Deterministic regex extractor — zero external dependency guarantee

Px**PyMuPDF**

Native PDF text extraction + page rendering for OCR

Oc**Tesseract OCR**

Automatic fallback for scanned/faxed pages

Tb**pdfplumber**

Table-aware text extraction for line-item bills

Tts**pyttsx3**

Offline text-to-speech — no cloud credential

Wa**Web Audio API**

Real-time amplitude analysis for avatar lip-sync

Sr**Web Speech API**

Browser-native microphone input for voice questions

Pt**Pytest**

76 automated backend tests, AI calls always mocked

Mem**In-Memory Repositories**

Repository pattern — ready to swap in a real database

8. Guardrails Reference

The single rule that repeats across every module in this app: **a model or engine may draft, but code always makes the final compliance-relevant decision.**

Source Verification

An AI-claimed value/citation must literally appear in the document text, or it is not trusted — extraction and chat both.

Agent Whitelist Filter

Growth Studio Discovery only ever returns agents from the domain's fixed whitelist — any other name is filtered and logged, never shown.

Confidence-Based Escalation

Below-threshold extraction confidence forces "Escalate to Human Review" and skips the AI simulation call entirely — in both the claim pipeline and Growth Studio.

Manager-Verified Protection

A human correction is marked verified and can never be silently overwritten by a later automatic re-extraction.

Missing-Field Overrides

No diagnosis forces a policy exception flag; no patient name forces human review — regardless of what the AI itself concluded.

Honest Failure Messages

"Not found," "unverified," and "AI unavailable" are three different, honestly-worded outcomes — never collapsed into one vague error.

9. Status & Badge Glossary

STATUS / BADGE	WHERE IT APPEARS	MEANING
Auto-Filled / Queued	Fax Intake, Dashboard	Extracted with no fields below the confidence threshold — ready for a quick approval pass
Needs Review	Fax Intake, Dashboard	At least one field fell below its confidence threshold, or the document match score was borderline
Resolved	Fax Intake, Dashboard	A manager has approved the claim
Invalid	Fax Intake, Dashboard	The uploaded document failed the relevance check and was rejected before extraction
Deleted	Fax Intake	Soft-deleted — excluded from normal views, restorable at any time
Manager Verified	Claim field badge	A human corrected this value; it is protected from future automatic overwrites
Conflict Detected	Claim field badge	This field read differently across two or more pages of the same document
HIGH / MEDIUM / LOW priority	Attention Required queue	Ranks which claims need a manager's attention first, computed from real confidence/status data
Auto-Approved	Growth Studio simulation	Every simulated agent found no issues and confidence was high enough to skip escalation
Human Review Required	Growth Studio simulation	A required field (e.g. patient name) was missing, forced in code regardless of the AI's own conclusion
Escalate to Human Review	Growth Studio simulation	Extraction confidence was below threshold — the AI simulation call was skipped entirely